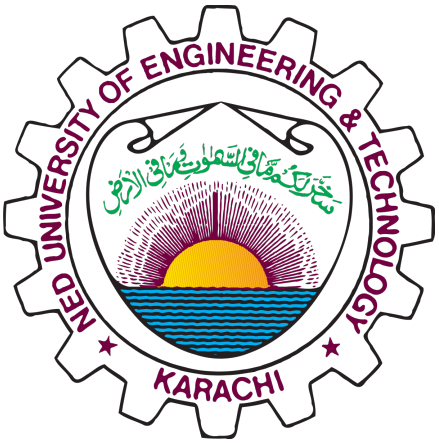


NED University
of Engineering and Technology

Team Reference Document

Team $(ttl)^3$

DP

0-1 Knapsack You are given n items with i_{th} item having weight w_i and value v_i. Given a knapsack with max Weight capacity W. Find the max value you can have.

Note that each item can only be taken **once**.

Time Complexity: $O(nW)$ **Space Complexity:** $O(W)$

```
int knapsack(int n, vector<int>& weights, vector<int>& values, int W) {
    vector<int> dp(W + 1, 0);
    for(int i = 0; i < n; i++) {
        for(int j = W; j >= weights[i]; --j) {
            dp[j] = max(dp[j], values[i] + dp[W - j]);
        }
    }
    return dp[W];
}
```

Complete Knapsack You are given n items with i_{th} item having weight w_i and value v_i. Given a knapsack with max Weight capacity W. Find the max value you can have.

Note that any item can be taken **unlimited times**.

Time Complexity: $O(nW)$ **Space Complexity:** $O(W)$

```
int knapsack(int n, vector<int>& weights, vector<int>& values, int W) {
    vector<int> dp(W + 1, 0);
    for(int i = 0; i < n; i++) {
        for(int j = weights[i]; j <= W; ++j) {
            dp[j] = max(dp[j], values[i] + dp[W - j]);
        }
    }
    return dp[W];
}
```

Multiple Knapsack You are given n items with i_{th} item having weight w_i and value v_i. Given a knapsack with max Weight capacity W. Find the max value you can have.

Note that each item can be taken k_i **times**.

Time Complexity: $O(n \sum \log k_i)$ **Space Complexity:** $O(W)$

```
// Just break each item in powers of 2 and treat each division as single item and apply 01 knapsack
```

Edit Distance Given two strings s_1 and s_2, find the minimum number of operations (insert, delete, replace) required to transform s_1 into s_2.

Time Complexity: $O(mn)$ **Space Complexity:** $O(mn)$

```
int editDistance(string& s1, string& s2) {
    int m = s1.size(), n = s2.size();
    vector<vector<int>> dp(m + 1, vector<int>(n + 1));
    for (int i = 0; i <= m; ++i) dp[i][0] = i;
    for (int j = 0; j <= n; ++j) dp[0][j] = j;
    for (int i = 1; i <= m; ++i) {
        for (int j = 1; j <= n; ++j) {
            dp[i][j] = min(dp[i - 1][j] + 1, dp[i][j - 1] + 1, dp[i - 1][j - 1] + (s1[i - 1] != s2[j - 1]));
        }
    }
    return dp[m][n];
}
```

Strings

Polynomial Hash

Time Complexity: $O(N)$ **Space Complexity:** $O(N)$

```
mt19937 rng(chrono::steady_clock::now().time_since_epoch().count());
const ll M = 991831889;
const ll C = uniform_int_distribution<ll>(0.1 * M, 0.9 * M)(rng);

struct HashString {
    int n;
    vector<ll> pows, sums;

    HashString(string s) : n(s.size()), pows(n + 1, 1), sums(n + 1) {
        for (int i = 1; i <= n; i++) {
            pows[i] = pows[i - 1] * C % M;
            sums[i] = (sums[i - 1] * C + s[i - 1]) % M;
        }
```

```
    }

    // Returns the hash of the substring [l, r)
    ll hash(int l, int r) {
        ll h = sums[r] - sums[l] * pows[r - l];
        return (h % M + M) % M;
    }
};
```

Trie

Time Complexity: $O(\sum |s_i|)$ **Space Complexity:** $O(26 * n)$

```
struct Vertex {
    int next[26];
    bool output = false;
    int link = -1;
    int p;
    char pch;
    int go[26];

    string text = "";
    int nearestOutput = -2;
    Vertex(int par = 0, char ch = '$'): p(par), pch(ch) {
        fill(begin(next), end(next), -1);
        fill(begin(go), end(go), -1);
    }
};
vector<Vertex> t(1);

int addString(string& s) {
    int v = 0;
    for(char ch: s) {
        int c = ch - 'a';
        if(t[v].next[c] == -1) {
            t[v].next[c] = t.size();
            t.emplace_back(v, ch);
        }
        v = t[v].next[c];
    }
    t[v].output = true;
    t[v].text = s;
    return v;
}

int go(int v, char ch);

int getLink(int v) {
    if(t[v].link == -1) {
        if(v == 0 || t[v].p == 0) {
            t[v].link = 0;
        } else {
            t[v].link = go(getLink(t[v].p), t[v].pch);
        }
    }
    return t[v].link;
}

int go(int v, char ch) {
    int c = ch - 'a';
    if(t[v].go[c] == -1) {
        if(t[v].next[c] != -1) {
            t[v].go[c] = t[v].next[c];
        } else {
            t[v].go[c] = v == 0 ? 0: go(getLink(v), ch);
        }
    }
    return t[v].go[c];
}

int getNearestOutput(int v) {
    if(t[v].nearestOutput == -2) {
        if(v == 0) {
            t[v].nearestOutput = -1;
        } else {
            int u = getLink(v);
            if(u == 0) {
                t[v].nearestOutput = -1;
```

```
            } else if(t[u].output) {
                t[v].nearestOutput = u;
            } else {
                t[v].nearestOutput = getNearestOutput(u);
            }
        }
    }
    return t[v].nearestOutput;
}
```

Z-Algorithm
Time Complexity: $O(N)$ **Space Complexity:** $O(N)$

```
vector<int> z_function(string& s) {
    int n = s.size();
    vector<int> z(n);
    int l = 0, r = 0;
    for(int i = 1; i < n; i++) {
        if(i < r) {
            z[i] = min(r - i, z[i - l]);
        }
        while(i + z[i] < n && s[z[i]] == s[i + z[i]]) {
            z[i]++;
        }
        if(i + z[i] > r) {
            l = i;
            r = i + z[i];
        }
    }
    return z;
}
```

Manacher Algorithm
Time Complexity: $O(N)$ **Space Complexity:** $O(N)$

```
#include "bits/stdc++.h"
using namespace std;

vector<int> manacher_odd(string s) {
    int n = s.size();
    s = "$" + s + "^";
    vector<int> p(n + 2, 0);
    int l = 0, r = 1;
    for (int i = 1; i <= n; i++) {
        if (i <= r)
            p[i] = min(r - i, p[l + (r - i)]);
        while (s[i - p[i]] == s[i + p[i]])
            p[i]++;
        if (i + p[i] > r) {
            l = i - p[i];
            r = i + p[i];
        }
    }
    return vector<int>(p.begin() + 1, p.end() - 1);
}

vector<int> manacher(const string& s) {
    string t;
    t.reserve(2 * s.size() + 1);
    for (char c : s) {
        t += '#';
        t += c;
    }
    t += '#';
    return manacher_odd(t);
}

void solve()
{
    string s;
    cin >> s;
    string org = s;

    vi p = manacher(s);
    vi ans(org.size());
    vi right(org.size());
    f(i, 1, p.size() - 1) {
```

```
        int len = p[i] - 1;
        int pos = i / 2 + len / 2;
        if(i % 2 == 0) pos--;
        // Starting from pos with 'len' length palindrome
        right[pos] = max(right[pos], len);
    }

    int maxLen = 0;
    fr(i, org.size() - 1, -1) {
        maxLen = max(right[i], maxLen);
        ans[i] = maxLen;
        maxLen -= 2;
    }

    f(i, 0, org.size()) {
        cout << ans[i] << ' ';
    }

}
```

KMP KMP Automation

Time Complexity: $O(N)\&O(N.M)$ **Space Complexity:** $O(N)\&O(N.M)$

```
#include "bits/stdc++.h"

using namespace std;

void kmpAutomation(int n, string& p) {
    int m = p.size();

    // Step 1: Compute KMP pi array
    vector<int> pi(m);
    for (int i = 1; i < m; i++) {
        int j = pi[i - 1];
        while (j > 0 && p[i] != p[j])
            j = pi[j - 1];
        if (p[i] == p[j])
            j++;
        pi[i] = j;
    }

    // Step 2: Build String Automaton
    vector<vector<int>> aut(m, vector<int>(26));
    for (int j = 0; j < m; j++) {
        for (int c = 0; c < 26; c++) {
            if (j > 0 && 'A' + c != p[j])
                aut[j][c] = aut[pi[j - 1]][c];
            else
                aut[j][c] = j + ('A' + c == p[j]);
        }
    }

    // Step 3: DP Table
    // dp[i][j] = ways to build a string of length i ending in automaton state j
    vector<vector<long long>> dp(n + 1, vector<long long>(m + 1, 0));
    dp[0][0] = 1;

    for (int i = 0; i < n; i++) {
        for (int j = 0; j <= m; j++) {
            if (dp[i][j] == 0) continue;

            if (j == m) {
                // Absorbing state: once matched, any character keeps it matched
                dp[i + 1][m] = (dp[i + 1][m] + dp[i][m] * 26LL) % MOD;
            } else {
                // Transition to next state for all 26 characters
                for (int c = 0; c < 26; c++) {
                    int nxt = aut[j][c];
                    dp[i + 1][nxt] = (dp[i + 1][nxt] + dp[i][j]) % MOD;
                }
            }
        }
    }

    cout << dp[n][m] << "\n";
}
```

DataStructures

Union Find

Time Complexity: $O(LogN)$ **Space Complexity:** $O(N)$

```
vi link(n + 1), size(n + 1);
f(i, 1, n + 1) link[i] = i;
f(i, 1, n + 1) size[i] = 1;

int find(int x) {
    if(x == link[x]) return x;
    return link[x] = find(link[x]);
}

bool same(int x, int y) {
    return find(x) == find(y);
}

void union(int a, int b) {
    a = find(a);
    b = find(b);
    if(a != b) {
        if(size[a] < size[b]) swap(a, b);
        size[a] += size[b];
        link[b] = a;
    }
}
```

Fenwick Tree

Time Complexity: $O(LogN)$ **Space Complexity:** $O(N)$

```
// Use 1 based indexing
long long sum(vll& tr, ll k) {
    ll s = 0;
    while(k >= 1) {
        s += tr[k];
        k -= k&-k;
    }
    return s;
}

void add(vll& tr, ll k, ll x) {
    while(k < tr.size()) {
        tr[k] += x;
        k += k&-k;
    }
}
```

Segment Tree

Time Complexity: $O(LogN)$ **Space Complexity:** $O(4 * N)$

```
// Set Identity / Base value returns
struct LazySegTree {
    int n;
    vector<ll> t, lz; // t = sum on segment, lz = pending add for children
    LazySegTree(int n) : n(n), t(4 * n), lz(4 * n) {}
    LazySegTree(const vector<ll>& a) : LazySegTree((int)a.size()) {
        build(1, 0, n, a);
    }
    void build(int v, int l, int r, const vector<ll>& a) {
        if (r - l == 1) { t[v] = a[l]; return; }
        int m = (l + r) / 2;
        build(2 * v, l, m, a);
        build(2 * v + 1, m, r, a);
        t[v] = t[2 * v] + t[2 * v + 1];
    }
    void apply(int v, int l, int r, ll x) { // apply +x to whole segment
        t[v] += x * (r - l);
        lz[v] += x;
    }
    void push(int v, int l, int r) {
        if (lz[v]) {
            int m = (l + r) / 2;
            apply(2 * v, l, m, lz[v]);
            apply(2 * v + 1, m, r, lz[v]);
            lz[v] = 0;
        }
```

```
    }
    void update(int v, int l, int r, int ql, int qr, ll x) {
        if (qr <= l || r <= ql) return;
        if (ql <= l && r <= qr) { apply(v, l, r, x); return; }
        push(v, l, r);
        int m = (l + r) / 2;
        update(2 * v, l, m, ql, qr, x);
        update(2 * v + 1, m, r, ql, qr, x);
        t[v] = t[2 * v] + t[2 * v + 1];
    }
    ll query(int v, int l, int r, int ql, int qr) {
        if (qr <= l || r <= ql) return 0; // identity
        if (ql <= l && r <= qr) return t[v];
        push(v, l, r);
        int m = (l + r) / 2;
        return query(2 * v, l, m, ql, qr) + query(2 * v + 1, m, r, ql, qr);
    }
    void update(int l, int r, ll x) { update(1, 0, n, l, r, x); } // a[l..r) += x
    ll query(int l, int r) { return query(1, 0, n, l, r); }  // sum a[l..r)
};
```

Graphs

Bellman Ford

Time Complexity: $O(N.M)$ **Space Complexity:** $O(N)$

```
for (int i = 1; i <= n; ++i) {
    for (auto [from, to, score] : edges) {
        if(distance[from] == INF) continue;
        ll new_distance = min(distance[to], distance[from] + score);
        if (i == n && new_distance < distance[to]) {
            cout << -1 << '\n'; // Negative Cycle
            return 0;
        }
        distance[to] = new_distance;
    }
}
cout << distance[n] << '\n';
```

Floyd Cycle

Time Complexity: $O(N)$ **Space Complexity:** $O(N)$

```
vector<int> extract_cycle(int x) {
    int a = x;
    int b = teleport[x];
    while (a != b) {
        a = teleport[a];
        b = teleport[teleport[b]];
    }
    vector<int> cycle;
    do {
        a = teleport[a];
        cycle.push_back(a);
    } while (a != b);
    return cycle;
}
```

SCC

Time Complexity: $O(N+M)$ **Space Complexity:** $O(N)$

```
vector<int> st;  // - stack holding the unclaimed vertices
vector<int> roots; // - keeps track of the SCC roots of the vertices
int timer;       // - dfs timestamp counter
vector<int> t_in; // - keeps track of the dfs timestamp of the vertices
vector<int> t_low; // - keeps track of the lowest t_in of unclaimed vertices
                   // reachable in the subtree

// implements the tarjan algorithm for strongly connected components
void dfs(int v, vector<vector<int>> const &adj, vector<vector<int>> &components) {

  t_low[v] = t_in[v] = timer++;
  st.push_back(v);

  for (auto u : adj[v]) {
    if (t_in[u] == -1) { // tree-edge
      dfs(u, adj, components);
```

```
      t_low[v] = min(t_low[v], t_low[u]);
    } else if (roots[u] == -1) { // back-edge, cross-edge or forward-edge to an unclaimed vertex
      t_low[v] = min(t_low[v], t_in[u]);
    }
  }

  if (t_low[v] == t_in[v]) { // vertex is a root
    components.push_back({v}); // initializes a new component with root v
    while (true) {
      int u = st.back();
      st.pop_back();
      roots[u] = v; // claims the vertex
      if (u == v)
        break;
      components.back().push_back(u); // adds vertex u to the component of v
    }
  }
}

// input: adj -- adjacency list of G
// output: components -- the strongy connected components in G
// output: adj_cond -- adjacency list of G^SCC (by root vertices)
void strongly_connected_components(vector<vector<int>> const &adj,
                                   vector<vector<int>> &components,
                                   vector<vector<int>> &adj_cond) {
  components.clear();
  adj_cond.clear();

  int n = adj.size();

  st.clear();
  roots.assign(n, -1);
  timer = 0;
  t_in.assign(n, -1);
  t_low.assign(n, -1);

  // applies the tarjan algorithm to all the vertices
  // adds vertices to the components in reverse topological order
  for (int v = 0; v < n; v++) {
    if (t_in[v] == -1) {
      dfs(v, adj, components);
    }
  }

  // adds edges to the condensation graph
  adj_cond.assign(n, {});
  for (int v = 0; v < n; v++) {
    for (auto u : adj[v])
      if (roots[v] != roots[u])
        adj_cond[roots[v]].push_back(roots[u]);
  }
}
```